

Alloy Verification Report

Title:

Complaint Management System



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Subject:

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1. Introduction

This report presents the Alloy-based structural verification of the Complaint Management System developed for the Software Verification and Validation (SVV) Lab project. The purpose of this verification is to formally model the system workflow, validate constraints, and analyze system behavior using Alloy Analyzer. The model verifies ticket workflow states, ticket assignment constraints, and system invariants.

2. Objectives

The objectives of Alloy verification are:

- To model the Complaint Management System using relational modeling
- To define system constraints and invariants
- To verify correctness of ticket workflow
- To perform counterexample analysis for invalid conditions

3. Relational Model

The Complaint Management System was modeled using Alloy signatures and relations.

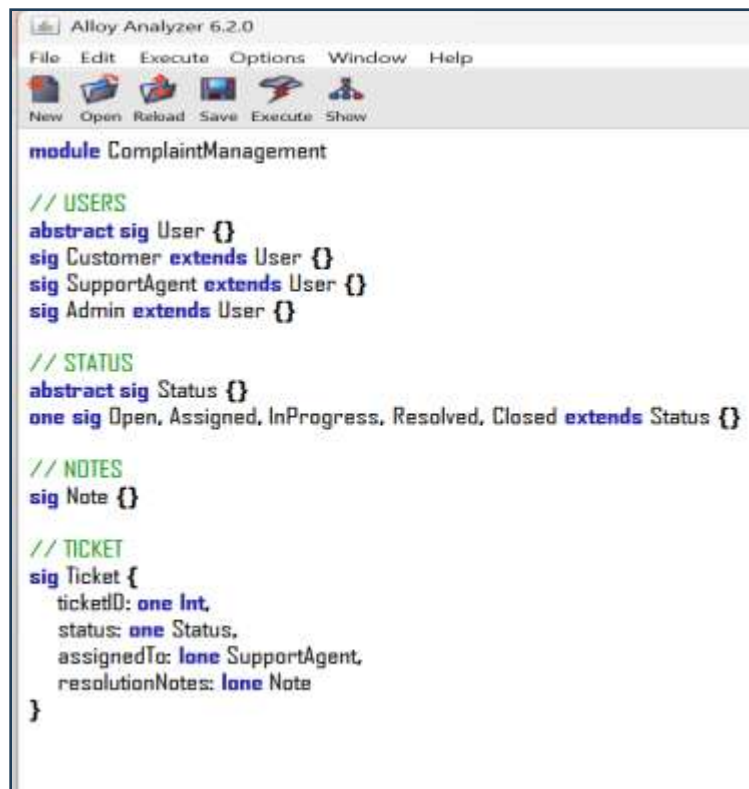


Figure 1 Relational model of the Complaint Management System in Alloy.

4. System Constraints

System constraints were implemented using Alloy facts. The following constraints were verified:

```
//Constraints
fact UniqueTicketIDs {
  all disj t1, t2: Ticket |
    t1.ticketID != t2.ticketID
}

fact ClosedTicketsCannotBeAssigned {
  all t: Ticket |
    t.status = Closed implies no t.assignedTo
}

fact ResolvedTicketsNeedNotes {
  all t: Ticket |
    t.status = Resolved implies some t.resolutionNotes
}
```

Line 74 Column 14 [modified]

Figure 2Alloy facts representing system constraints and invariants.

5. Constraint Verification

Assertions were checked using Alloy Analyzer to verify system correctness. The following assertions were verified successfully:

- NoDuplicateIDs
- ClosedTicketNoAgent

```
Executing "Check NoDuplicateIDs for 5"
Actual scopes: 5 User, 5 Customer, 5 SupportAgent, 5 Admin, 5 Status, exactly 1 Open, exactly 1 Assigned, exactly 1 InProgress, exactly 1 Resolved, exactly 1 Closed, 5 Note, 5 Ticket
Solver=sat4j Bitwidth=4 MaxSeq=5 SkolemDepth=1 Symmetry=20 Mode=batch
2205 vars, 190 primary vars, 4245 clauses, 201ms.
No counterexample found. Assertion may be valid. 44ms.
```

Figure 3Successful verification results showing no counterexamples found.

Executing "Check NoDuplicateIDs for 5"

Actual scopes: 5 User, 5 Customer, 5 SupportAgent, 5 Admin, 5 Status, exactly 1 Open, exactly 1 Assigned, exactly 1 InProgress, exactly 1 Resolved, exactly 1 Closed, 5 Note, 5 Ticket
Solver=sat4j Bitwidth=4 MaxSeq=5 SkolemDepth=1 Symmetry=20 Mode=batch
2205 vars, 190 primary vars, 4245 clauses, 20lms.
No counterexample found. Assertion may be valid. 44ms.

Figure 4 Successful verification results showing no counterexamples found.

6. Model Execution

The Alloy model was executed successfully using the run command. The generated instance graph visually represents the relationships between tickets, users, support agents, and ticket states. The execution confirmed that the model is structurally consistent.

Executing "Run run\$3 for 5"

Actual scopes: 5 User, 5 Customer, 5 SupportAgent, 5 Admin, 5 Status, exactly 1 Open, exactly 1 Assigned, exactly 1 InProgress, exactly 1 Resolved, exactly 1 Closed, 5 Note, 5 Ticket
Solver=sat4j Bitwidth=4 MaxSeq=5 SkolemDepth=1 Symmetry=20 Mode=batch
1897 vars, 180 primary vars, 3374 clauses, 92ms.
Instance found. Predicate is consistent. 42ms.

Figure 5 instance found

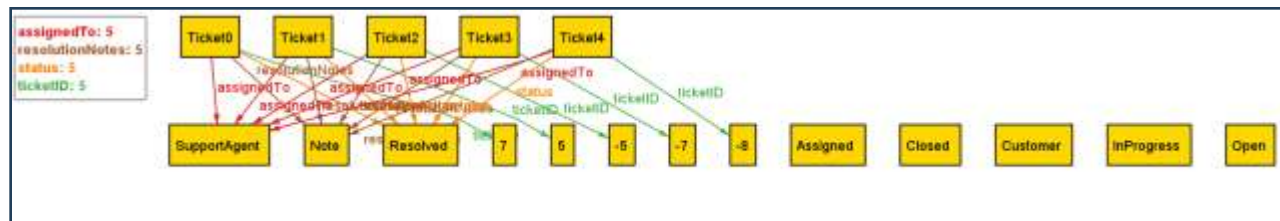


Figure 6 Alloy-generated instance graph of the Complaint Management System.

7. Counterexample Analysis

A counterexample analysis was performed by intentionally adding an incorrect assertion named WrongRule. The incorrect assertion stated that closed tickets must always have assigned support agents. This condition violates the actual workflow constraint that closed tickets should not remain assigned.

The Alloy Analyzer successfully detected the invalid condition and generated a counterexample.

Executing "Check WrongRule for 5"

Actual scopes: 5 User, 5 Customer, 5 SupportAgent, 5 Admin, 5 Status, exactly 1 Open, exactly 1 Assigned, exactly 1 InProgress, exactly 1 Resolved, exactly 1 Closed, 5 Note, 5 Ticket
Solver=sat4j Bitwidth=4 MaxSeq=5 SkolemDepth=1 Symmetry=20 Mode=batch
1999 vars, 185 primary vars, 3550 clauses, 79ms.
Counterexample found. Assertion is invalid. 27ms.

Figure 8 counter example found

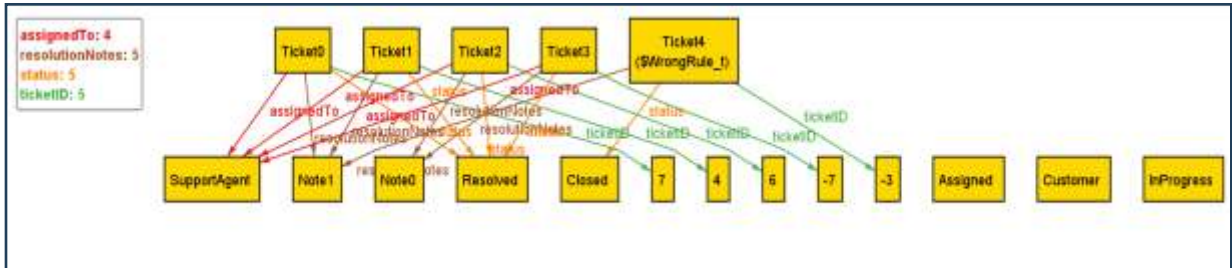


Figure 7 Counterexample generated for the invalid assertion WrongRule.

8. Results

The Alloy model successfully verified the structural correctness of the Complaint Management System. All important constraints and workflow rules were validated successfully.

The counterexample analysis also demonstrated Alloy's capability to identify invalid or inconsistent system behavior.

9. Conclusion

The Alloy-based verification process improved confidence in the correctness and consistency of the Complaint Management System.

The project successfully demonstrated the use of formal verification techniques for validating constraints, analyzing workflow behavior, and detecting inconsistencies through counterexample analysis.